Program: **Diploma in Electrical Engineering**	
Course Code: **5541A**	Course Title: **Introduction to Electric Vehicles**
Semester: **5**	Credits: **4**
Course Category: **Engineering Science**	
Periods per week: **4 (L:3 T:1 P:0)**	Periods per semester: **60**

Course Objectives:

- To know the history, development, classification, and basic terminology of automobile vehicles.
- Developmentof electric vehicles and appreciate the use EVs.
- To study batteries used in Electric Vehicle.
- To recognize lighting, heating and chemical effects of electric current.

Course Prerequisites:

Topic	Course code	Course Name	Semester
Knowledge of basic science	1003 & 2003	Applied Physics I& II	1,2
Basic Concept of Electro Chemistry	1004	Applied Chemistry	2

CourseOutcomes:

Oncompletionofthecourse,thestudentwillbeableto:

COs	Description	Duration (Hours)	Cognitive Level
CO1	Summarize different classification of vehicles and terminology of Chassis specifications.	15	Understanding
CO2	Appreciat the use of electric vehicles and classification of storage batteries.	15	Applying
CO3	Discuss the construction and working of batteries Used in vehicles.	14	Understanding

CO4	Summarize the luminous, heating, and chemical effects of electric current.	14	Understanding
	SeriesTest	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	2						
CO3	2						
CO4	3						

3-Strongly mapped,2-Moderately mapped,1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	CognitiveLevel
CO1	**Summarize different classification of vehicles and terminology of Chassis specification.**		
M1.01	Explain the History of Automobile Vehicles	3	Understanding
M1.02	Outline the various classifications of Vehicles	2	Understanding
M1.03	Outline different Vehicle layouts and explain various terminology of vehicle specification	6	Understanding
M1.04	Understand the working principle of petrol & diesel engines and the environmental impact of fossil fueled engines.	4	Understanding

Contents:

HistoryandClassification

Automobile Vehicles - Definition – History.

Classification of Automobile – Major systems of an Automobile. Chassis - definition, chassis layout of two, three & four – wheeler with major components – definition of: wheelbase - wheel track - overall length - front overhung - rear overhung height of CG point – ground clearance – gross weight and kerb weight.

Heatengines - Definition - types -engine terminology- classification of IC engineswith respect to different parameters, Two stroke & four stroke SI engines- working,Two stroke & Four strokeCIengines-working, comparison of SIand CIengines -comparison of Two stroke and Four stroke engines - emission characteristics of diesel and petrol engines

CO2	**Appreciate the use of electric vehicles.**		
M2.01	Explain the back ground & development of electric vehicles.	3	Understanding
M2.02	List the various types of electric vehicles and their components and draw the layoutofdifferent electric vehicles.	6	Applying
M2.03	Classify Storage batteries and summarize various terms related.	3	Understanding
M2.04	Identify the parts of lead acid batteries and its care and maintenance.	3	Understanding
	SeriesTest-I	1	

Contents:

Development of EVs:

Development of electric vehicles in India and abroad - types of EVs-components-layout-merits of EVs-comparison between conventional and electric vehicles.

Contents Storage battery- cell – battery - primary and secondary batteries - battery ratings –efficiency - ampere hour efficiency - watt hour efficiency - lead acid batteries-identification of parts (brief idea only) - electrical Specifications – voltage – capacity - factors affecting capacity - indications of a fully charged cell - applications - care and maintenance – lithium -ion battery - basic idea – applications - series and parallel connections of batteries - list the methods of charging.

CO3	**Discuss the construction and working of batteries used in vehicles.**		
M3.01	Identify the various types of batteries used in EVs	3	Understanding
M3.02	Comparison of batteries used in EV systems	6	Understanding
M3.03	Importance of ultra-capacitors in EV systems.	2	Understanding
M3.04	Identify the methods of earthing in EV systems	3	Understanding

Contents:

Electro chemical batteries: Lead Acid Batteries – Nickel based batteries – Lithium based batteries - Fuel Cell. Construction and working of lithium-based Batteries- Advantages of Lithiumbatteries.
Comparison of batteries with respect to specific energy, specificpower, cyclelife, cost.
Brief introduction of Ultra Capacitors-Importance of ultracapacitors in hybridization of energy storage devices- Advantages
Types of earthing systems-TT earthing, TN earthing, GFCI protection for EV Charging stations.

CO4	**Summarize the luminous, heating, and chemical effects of electric current.**		
M4.01	Explain the basic concepts of various lamps for residential and commercial Purposes	3	Understanding
M4.02	Summarize the basic principle of electric heating and modes of heat transfer and classify various methods of electric heating.	6	Understanding
M4.03	Summarize the basic principle of electrolysis and familiarize the applications of electrolysis	2	Understanding
M4.04	Classify the electric and magnetic materials	3	Understanding
	Series Test- II	1	

Contents:

Electrical Lamps–types–basic diagram –working–incandescent–fluorescent–high pressure & low - pressure sodium vapour - LED -compact fluorescent – induction lamp

Electric Heating- Basic principle - advantages - requirements of good heating element - modes of heat transfer- methods of electric heating – principle -operation– resistance heating – direct – indirect – induction – direct core–dielectric (principle of operation only).

Electrolysis–basic principle-Faraday’s laws-brief idea of applications of electrolysis.

Electric materials–classification–properties-examples-conducting materials-insulating **Magnetic materials**- Ferromagnetic - Diamagnetic- Paramagnetic (properties & examples only)

Text/Reference

T/R	BookTitle/Author
T1	Ehsani, M. Modern Electric, Hybrid Electricand Fuel Cell Vehicles: Fundamentals, Theory and Design, CRC Press
T2	Electric vehicle Technology-Prof. Sunil.R. Pawar-Notion press publication.
T3	Electric vehicle engineering-PerEnge,NickEnge,StephenZoepf-McGrawHill.
T4	Automobile Engg.-Vol-II-DrKirpalSingh-StandardPublishersDistributors.
T5	Theraja,B.L.,ElectricalTechnologyVol–I,S.ChandPublications,New Delhi, 2015, ISBN: 9788121924405
T6	Theraja,B.L.;Theraja,A.K.,ATextBookofElectricalTechnologyVol-III, S.ChandandCo.Ramnagar,NewDelhi,ISBN:9788121924900
R1	V.K.Mehta,RohitMehtaBasicElectricalEngineeringS.ChandPublications, New Delhi,
R2	Jegathesan,V.,BasicElectricalandElectronicsEngineering,WileyIndia,New Delhi, 2015, ISBN : 97881236529513
R3	ChanC.C.andK.T.Chau,ModernElectricVehicleTechnology,Oxford Science Publication,
R4	LechnerG.andH.Naunheimer,AutomotiveTransmissions:Fundamentals, Selection,Design and Application, Springer
R5	Rashid,M.H.PowerElectronics:Circuits,DevicesandApplications,3rd edition, Pearson,
R6	Moorthi,V.R.PowerElectronics:Devices,CircuitsandIndustrial Applications, OxfordUniversity Press
R7	Krishnan,R.Electricmotordrives:modelling,analysis,andcontrol,Prentice Hall11. Krause, O. P. ; C. Wasynczuk, S. D. Sudhoff, Analysis of electric machinery, IEEE Press

OnlineResources

Sl.No.	WebsiteLink
1	https://nptel.ac.in/courses/108105112/
2	https://nptel.ac.in/courses
3	https://nptel.ac.in/courses/108/103/108103009/
4	https://onlinecourses.nptel.ac.in/noc20_ee99/preview
5	https://energyeducation.ca/encyclopedia/Main_Page

6	https://www.evacad.com
7	https://www.youtube.com/watch?v=UgtjRob5qMg
8	https://youtu.be/3E1SXG7VkQk?si=grGF9k7DbWxkHatQ